

Operations on Sets

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7.1 The basic operations

Throughout, sets live inside a fixed universal set U .

Definition 7.1 (Union, intersection, difference, complement).

$$\begin{aligned} A \cup B &= \{x \mid x \in A \vee x \in B\} && \text{(union)} \\ A \cap B &= \{x \mid x \in A \wedge x \in B\} && \text{(intersection)} \\ A \setminus B &= \{x \mid x \in A \wedge x \notin B\} && \text{(difference)} \\ \overline{A} &= U \setminus A = \{x \in U \mid x \notin A\} && \text{(complement)} \\ A \oplus B &= (A \setminus B) \cup (B \setminus A) && \text{(symmetric difference).} \end{aligned}$$

Two sets with $A \cap B = \emptyset$ are called *disjoint*.

Each set operation mirrors a logical connective: $\cup \leftrightarrow \vee$, $\cap \leftrightarrow \wedge$, $\overline{} \leftrightarrow \neg$, $\oplus \leftrightarrow \oplus$. This dictionary transfers every law of Chapter 2 into a set identity.

Example 7.2. With $U = \{1, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$: $A \cup B = \{1, \dots, 7\}$, $A \cap B = \{4, 5\}$, $A \setminus B = \{1, 2, 3\}$, $\overline{B} = \{1, 2, 3, 8, 9, 10\}$, $A \oplus B = \{1, 2, 3, 6, 7\}$.

Theorem 7.3 (Inclusion–exclusion, basic form). For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$.

Example 7.4 (Combining operations). With the sets above, compute $\overline{A \cap B} \cap A$: $A \cap B = \{4, 5\}$, so $\overline{A \cap B} = U \setminus \{4, 5\}$, and intersecting with A leaves $\{1, 2, 3\} = A \setminus B$. In fact $\overline{A \cap B} \cap A = A \setminus B$ always — provable by the identities below.

Example 7.5 (Empty regions). $A \cap \overline{A} = \emptyset$ and $(A \setminus B) \cap (B \setminus A) = \emptyset$ for any A, B : some Venn regions are empty by pure logic, regardless of the particular sets.

7.2 Set identities

Theorem 7.6 (Set identities). For all $A, B, C \subseteq U$:

Identity	$A \cup \emptyset = A, \quad A \cap U = A$
Domination	$A \cup U = U, \quad A \cap \emptyset = \emptyset$
Idempotence	$A \cup A = A, \quad A \cap A = A$
Double complement	$\overline{\overline{A}} = A$
Commutativity	$A \cup B = B \cup A, \quad A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$, and dually
Distributivity	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, and dually
De Morgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{A \cap B} = \overline{A} \cup \overline{B}$
Absorption	$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$
Complement	$A \cup \overline{A} = U, \quad A \cap \overline{A} = \emptyset$

Also $A \setminus B = A \cap \overline{B}$, the workhorse for eliminating differences.

7.3 Proving identities

Three standard methods.

1. The element method (double inclusion). To prove $X = Y$, show $x \in X \Rightarrow x \in Y$ and conversely, unfolding the definitions into logic and applying the propositional laws.

Example 7.7. Prove $\overline{A \cup B} = \overline{A} \cap \overline{B}$:

$$x \in \overline{A \cup B} \iff \neg(x \in A \vee x \in B) \iff (x \notin A) \wedge (x \notin B) \iff x \in \overline{A} \cap \overline{B},$$

the middle step being De Morgan's law for propositions.

2. Algebraic manipulation. Chain identities from Theorem 7.6, replacing equals by equals — as in Example 7.4: $\overline{A \cap B} \cap A = (\overline{A \cup B}) \cap A = (\overline{A} \cap \overline{B}) \cap A = \emptyset \cup (A \cap \overline{B}) = A \setminus B$.

3. Membership tables. Analogous to truth tables: each row records whether x belongs (1) or not (0) to each set; two expressions are equal iff their columns coincide.

Example 7.8 (Membership-table proof). $A \oplus B = (A \cup B) \setminus (A \cap B)$:

A	B	$A \cup B$	$A \cap B$	$(A \cup B) \setminus (A \cap B)$	$A \oplus B$
1	1	1	1	0	0
1	0	1	0	1	1
0	1	1	0	1	1
0	0	0	0	0	0

The last two columns agree, proving the identity.

A membership table over n set letters has 2^n rows — it is exactly a truth table in disguise, so the method is complete for identities built from $\cup, \cap, \setminus, \oplus$ and complement.

7.4 Generalised unions and intersections

For a family $A_1, \dots, A_n$ (or an infinite family):

$$\bigcup_{i=1}^n A_i = \{x \mid \exists i, x \in A_i\}, \quad \bigcap_{i=1}^n A_i = \{x \mid \forall i, x \in A_i\}.$$

Example 7.9. With $A_i = \{i, i+1, i+2, \dots\} \subseteq \mathbb{N}$: $\bigcup_{i \geq 1} A_i = \mathbb{N}$ and $\bigcap_{i \geq 1} A_i = \emptyset$ (no natural number belongs to every A_i).

7.5 Summary

Union, intersection, difference, complement and symmetric difference mirror $\vee, \wedge, \setminus, \neg, \oplus$; consequently the identity, domination, idempotent, distributive, De Morgan, absorption and complement laws all transfer from logic to sets. Identities are proved by the element method, by algebraic chaining, or mechanically by membership tables.

Exercises

Exercise 7.1 EASY

Let $U = \{1, \dots, 12\}$, $A = \{1, 3, 5, 7, 9, 11\}$, $B = \{3, 6, 9, 12\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$, $B \setminus A$, $\overline{A}$ and $A \oplus B$.

Exercise 7.2 EASY

Draw (or describe region by region) the Venn diagram of $(A \cup B) \setminus (A \cap B)$ and name the operation it represents.

Exercise 7.3 EASY

Two clubs have 28 and 19 members; 7 people belong to both. How many people belong to at least one club?

Exercise 7.4 MEDIUM

Prove by the element method: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Exercise 7.5 MEDIUM

Prove algebraically (citing identities): $A \cup (B \setminus A) = A \cup B$.

Exercise 7.6 MEDIUM

Verify by a membership table: $A \setminus (A \setminus B) = A \cap B$.

Exercise 7.7 MEDIUM

Show that $A \subseteq B$ if and only if $A \cup B = B$, and if and only if $A \cap B = A$.

Exercise 7.8 HARD

Prove that symmetric difference is associative: $(A \oplus B) \oplus C = A \oplus (B \oplus C)$. *Hint: a membership table over three letters, or the parity characterisation.*

Exercise 7.9 HARD

Solve for X : the equation $A \oplus X = B$ has exactly one solution for given $A, B \subseteq U$. Find it and prove uniqueness.

Exercise 7.10 HARD

Prove the three-set inclusion–exclusion formula:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Solutions to the Exercises

Solution 7.1. $A \cup B = \{1, 3, 5, 6, 7, 9, 11, 12\}$; $A \cap B = \{3, 9\}$; $A \setminus B = \{1, 5, 7, 11\}$; $B \setminus A = \{6, 12\}$; $\overline{A} = \{2, 4, 6, 8, 10, 12\}$; $A \oplus B = \{1, 5, 6, 7, 11, 12\}$. ■

Solution 7.2. The region consists of the two “lune” parts of the discs — everything in exactly one of A, B , excluding the central lens $A \cap B$ and the exterior. This is the symmetric difference $A \oplus B$. ■

Solution 7.3. $|A \cup B| = 28 + 19 - 7 = 40$. ■

Solution 7.4. $x \in A \setminus (B \cup C) \iff x \in A \wedge \neg(x \in B \vee x \in C) \iff x \in A \wedge x \notin B \wedge x \notin C$ (De Morgan). Also $x \in (A \setminus B) \cap (A \setminus C) \iff (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C)$, which by idempotence of $\wedge$ is the same condition. Hence the sets are equal. ■

Solution 7.5. $A \cup (B \setminus A) = A \cup (B \cap \overline{A}) = (A \cup B) \cap (A \cup \overline{A}) = (A \cup B) \cap U = A \cup B$, using difference-as-intersection, distributivity, complement and identity laws. ■

Solution 7.6.

A	B	$A \setminus B$	$A \setminus (A \setminus B)$	$A \cap B$
1	1	0	1	1
1	0	1	0	0
0	1	0	0	0
0	0	0	0	0

Columns 4 and 5 coincide. ■

Solution 7.7. If $A \subseteq B$: every element of $A \cup B$ is in B , so $A \cup B \subseteq B \subseteq A \cup B$, giving equality; and $A \subseteq A \cap B \subseteq A$ similarly. Conversely, if $A \cup B = B$ then $A \subseteq A \cup B = B$; if $A \cap B = A$ then $A = A \cap B \subseteq B$. ■

Solution 7.8. Membership in a symmetric difference holds exactly when membership holds in an *odd* number of the operands: for two sets this is the definition. Then $x \in (A \oplus B) \oplus C$ iff x lies in an odd number of $\{A, B, C\}$ — check: it requires oddness between $(A \oplus B)$ and C , and $x \in A \oplus B$ records oddness between A and B ; combining parities gives overall parity. The same characterisation holds for $A \oplus (B \oplus C)$, so the two sides are equal. (The eight-row membership table gives the same conclusion mechanically.) ■

Solution 7.9. Take $X = A \oplus B$. Then, using associativity (Exercise 7.8), $A \oplus (A \oplus B) = (A \oplus A) \oplus B = \emptyset \oplus B = B$. Uniqueness: if $A \oplus X = A \oplus Y$, then XOR-ing both sides with A gives $X = Y$. (The structure $(\mathcal{P}(U), \oplus)$ is a group in which every element is its own inverse.) ■

Solution 7.10. Count the contribution of an element x belonging to exactly $k \geq 1$ of the three sets. Left side counts x once. Right side counts it $\binom{k}{1} - \binom{k}{2} + \binom{k}{3}$ times: for $k = 1$: 1; $k = 2$: $2 - 1 = 1$; $k = 3$: $3 - 3 + 1 = 1$. Every element is counted once on each side, so the totals agree. (Chapter 10 generalises this to n sets.) ■